

Kenneth Plum Toft

Copenhagen, Denmark

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Summary

Highly motivated MSc Computer Science candidate with specialized expertise in AI, Algorithms, and Full-Stack Engineering, driven by a passion for translating cutting-edge AI research into production-grade healthcare and enterprise solutions. Proven track record of taking AI-centric use-cases from ideation to deployment, including the development of GenAI platforms, RAG/GraphRAG systems, and advanced speech recognition technologies. Adept at architecting cloud-native, scalable applications using Python, C#, and React/TypeScript, while orchestrating robust CI/CD pipelines and containerized infrastructure on AWS. Brings valuable cross-functional experience collaborating with data scientists, domain experts, and product stakeholders to optimize AI workflows, ensure highest-quality outputs, and drive business impact. Combines deep technical proficiency in frontier AI technologies with strong agile development practices, continuous learning mindset, and excellent communication skills to thrive in dynamic, multinational tech environments.

Work Experience

- Dictus** Copenhagen, Denmark
Software Developer Aug 2023 - Now
 - Spearheaded end-to-end GenAI feature development from ideation to production, integrating frontier LLMs (DanskGPT, Gemma) with advanced prompt engineering and pre/post model processing pipelines.
 - Architected cloud-native, scalable full-stack applications using C#/.NET, React/TypeScript, and Python-based AI microservices to support enterprise-grade digital dictation and transcription workflows.
 - Deployed state-of-the-art speech AI models (Whisper, Wav2Vec, TitaNet-L) for real-time healthcare transcription and speaker diarization, optimizing model performance for high-fidelity audio data.
 - Engineered robust CI/CD pipelines and containerized deployments (Docker, Jenkins, Git) on AWS/cloud infrastructure to ensure high availability, rapid iteration, and seamless integration of AI components.
 - Collaborated closely with data scientists, domain experts, and product teams to translate AI innovations into tangible business solutions, providing continuous feedback and evaluation benchmarks for AI products.
 - Developed and maintained large-scale applications for national parliamentary clients (Stortinget), demonstrating adaptability and excellence in complex, regulated environments.
- Dictus** Copenhagen, Denmark
Software Developer Intern Aug 2022 - Aug 2023
 - Developed and deployed full-stack web platforms for AI-driven audio data crowdsourcing, streamlining client data collection and model training workflows.
 - Engineered a commercial-grade AI application integrating Automatic Speech Recognition (ASR) and deep learning-based speaker separation, demonstrating rapid prototyping and production readiness.

- Collaborated with engineering teams to optimize backend architectures using C#, Python, and modern frontend frameworks, ensuring scalability and maintainability.
- Applied modern software development practices and cross-functional teamwork to deliver high-quality, user-centric AI features within agile environments.

- Greenwood Engineering** Copenhagen, Denmark
Student Worker Aug 2021 - Sep 2022
 - Managed enterprise IT infrastructure, including server provisioning, virtual machine orchestration, and maintenance of CI/CD and collaboration toolchains (Git, Jenkins, Wiki).
 - Provided comprehensive technical support and system optimization for engineering teams, ensuring operational continuity and efficient resource allocation.
 - Demonstrated strong problem-solving skills and adaptability in a fast-paced manufacturing and infrastructure assessment environment.
- Tradir.io** Copenhagen, Denmark
Software Engineer Dec 2020 - Aug 2021
 - Built scalable CRM backend services in an agile environment using Python, Django-REST, AWS, and PostgreSQL, directly supporting global export workflow optimization.
 - Integrated third-party AI/automation APIs (Nylon, Mailgun, Nanonets) to streamline data processing and client communications, enhancing platform functionality.
 - Collaborated with multidisciplinary teams to deliver high-quality software solutions, adhering to best practices in API design, cloud deployment, and agile development.
- Widex** Copenhagen, Denmark
Firmware Updater Feb 2019 - Oct 2019
 - Executed precision firmware updates for medical hearing devices, adhering to strict quality, safety, and regulatory standards in the medtech sector.
 - Ensured reliable device performance and maintained meticulous documentation in compliance with healthcare technology protocols.
- Power** Copenhagen, Denmark
IT-Support Sep 2016 - Oct 2017
 - Delivered frontline IT support and data recovery services across a wide range of consumer electronics and mobile devices.
 - Honed customer-centric troubleshooting, technical communication, and rapid problem-resolution skills in a high-volume retail environment.

Projects

Master's Thesis: Synergy Effects of GraphRAG, Fine-tuned Embeddings, and Specialized LLMs on Automated Content

- Generation**
Lead Researcher & Developer 2024 - 2026
 - Developing a comprehensive full-stack application to benchmark automated content generation quality against standard AI baselines.
 - Implementing advanced RAG and GraphRAG architectures, leveraging Knowledge Graphs combined with Retrieval Augmented Generation for contextual accuracy.
 - Fine-tuning and deploying specialized Large Language Models, directly addressing challenges in optimizing model inference costs and selecting technical frameworks for high-fidelity generative outputs.

- Designing evaluation metrics and feedback loops to support AI process re-imagination and platform generalization for enterprise deployment.

AI-Powered Audio Transcription & Speaker Separation

- **Platform**

Lead Developer & AI Architect 2022 - 2023

- Built a commercial-grade full-stack application integrating Automatic Speech Recognition (ASR) and deep learning-based speaker diarization models.
- Architected end-to-end data pipelines for audio processing, model inference, and user-facing web interfaces using Python, C#, and React.
- Project successfully transitioned from academic research to commercial development, demonstrating rapid prototyping, production readiness, and market viability.

Education

Technical University of Denmark Copenhagen, Denmark

- *MSc, Computer Science and Engineering (Specialization in AI and Algorithms)* 2024 - Now

Hanyang University (South Korea) Copenhagen, Denmark

- *BSc, Computer Science (Exchange)* 2020 - 2021

Technical University of Denmark Copenhagen, Denmark

- *BSc, Computer Engineering* 2019 - 2023

Technical Skills

Python, C#, JavaScript, TypeScript, SQL, Java, C, ASP.NET, Django REST, React, Node.js, RabbitMQ, PyTorch, HuggingFace, FastAPI, AWS, Google Cloud Platform, DVC, Weights & Biases, PostgreSQL, MySQL, Docker, Jenkins, Git, GitHub, AI/LLM Integration, Prompt Engineering, RAG, GraphRAG, Vector Databases, Semantic Search, Speech Recognition, Speaker Diarization, Model Fine-tuning, CI/CD Pipelines, Cloud-Native Architecture, Agile Development, Rapid Prototyping, Cross-Functional Collaboration, Stakeholder Communication, Technical Advisory, AI Evaluation & Benchmarking, Healthcare Domain AI, Enterprise Software Architecture, Modern AI Tooling & IDEs, AI Process Optimization

Languages

- Danish:** Native
- English:** Professional Working Proficiency
- Korean:** Low Intermediate